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Implementation and enforcement in the United Kingdom

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12.1 Background

UK government initiatives to encourage the implementation of HACCP-based food safety control systems began in the early 1990s. This reflected the growing international recognition that HACCP was the management system of choice to ensure food safety and consumer protection. The initiatives also responded to key recommendations made by a UK Expert Committee which examined the rise in UK food poisoning during the 1980s. With regard to HACCP, the Committee Report[1] recommended that:

- food processes should be designed on HACCP principles
- enforcement staff should seek to encourage HACCP through their inspection activities
- enforcement staff should receive appropriate training in HACCP systems and their application.

Early initiatives focused on increasing the general awareness of HACCP and ensuring that appropriate training courses and materials were available to both the food industry and enforcement officers. For example, the professional Core Curriculum for local government environmental health officers (who constitute the major part of the UK Food Safety Inspectorate) was examined to ensure that it contained enough practical training to enable inspectors to carry out assessments of HACCP-based systems. A nation-wide programme of training and awareness seminars for both industry and enforcement officials was also carried out and a range of HACCP materials, training aids and packages were developed or encouraged to help facilitate HACCP implementation in food businesses.

Some parts of the UK food industry, particularly the larger manufacturers, were quick to implement HACCP management systems and such systems were required within commercial supplier specifications and contracts. Also, companies with HACCP in place, particularly within ISO 9000 (quality control systems) were more able to sustain 'due diligence' defences available in UK law from 1990 in the event of a food safety problem.[2] As such, enforcement authorities, even in the early 1990s, were increasingly being required to assess the adequacy of HACCP systems on a legal basis where these systems were being put forward by businesses as part of a due diligence defence. However, enforcement officers also had a key advisory role with smaller businesses during this period, increasing their awareness of HACCP and encouraging them to put HACCP-based controls in place.

12.2 UK legal requirements for HACCP

Since 1993, as part of the European legislative framework to support the single market, HACCP-based requirements have been progressively introduced into UK law through the implementation of a range of EU food hygiene directives.

The Hygiene of Foodstuffs Directive 93/43EEC (covering food retailers, caterers and manufacturers of non-animal origin products), for example, requires certain principles of HACCP to be applied in these food businesses. Although the current legislation does not include an explicit requirement to document the system, in line with government advice, enforcement officials do encourage appropriate documentation and record keeping.

Separate European product specific legislation governs establishments manufacturing products of animal origin, i.e meat products, fish products, egg products and milk products. With the exception of fresh meat and milk, the legislation requires HACCP-based 'Own Checks' controls to be in place. These 'Own Checks' controls do require establishments to document procedures, such as the identification of critical points, monitoring arrangements and sampling regimes. Individual establishments also need to be approved by the authorities in order to trade.

12.3 The regulatory authorities

Historically, UK government responsibilities for food safety and hygiene have been shared between the Ministry of Agriculture, Fisheries and Food (MAFF) and the Department of Health. Between them, these two government departments developed and implemented the UK food policy and legislation including policy related to HACCP. On 1 April 2000 a UK Food Standards Agency was established which radically altered these arrangements and the Agency now has the UK policy and legislative responsibilities for food safety and food standards from farm to fork. The Food Standards Agency is a non-ministerial government

department, governed by an independent 14-member board, accountable to Parliament through health ministers.

However, the day-to-day enforcement of food safety in the UK, including the enforcement of legal requirements relating to HACCP, is in the main discharged by local environmental health officials working in approximately 500 local authorities. Key exceptions to these arrangements relate to enforcement work within the fresh meat sector such as slaughterhouses and cutting plants, and on-farm dairy control. These sectors fall to the Food Standards Agency's Meat Hygiene Service and Dairy Hygiene Inspectorate respectively. However, currently there are no requirements for HACCP in the EU legislation applicable to these two sectors.

Environmental health officials have legal powers to:

- enter food businesses at all reasonable hours, if need be by force
- serve official notices to secure improvement and regulatory compliance
- close down food businesses immediately where there is imminent risk to public health
- prosecute proprietors or individuals for breaches of food safety and hygiene legislation
- detain and/or seize food failing to meet food safety requirements.

The activities of local enforcement officials are guided by statutory codes of practice made under the Food Safety Act and issued by central government. These codes contain advice on enforcement approach; recommended inspection frequencies; the nature of inspections to be carried out; and guidance on assessing HACCP-based requirements. These official codes are also supplemented by guidance issued by the Local Authority Co-ordinating Body for Food and Trading Standards (LACOTS).

In the UK food establishments are invariably subject to commercial auditing associated with supplier requirements. This reflects the very high percentage of UK food retailed through large supermarkets, which routinely assess supplier standards. Although the scope of these audits can cover other issues, e.g. quality, they usually incorporate assessments of the HACCP food safety arrangements, particularly when the audit is carried out within a recognised accreditation framework. Whilst such audits are not directly related to enforcement activities, they make an increasingly valuable contribution to food safety. Government enforcement codes of practice also allow commercial audits, particularly by accredited third parties, to be taken into account by local enforcement officials when considering the nature and frequency of their inspections.

12.4 Take-up of HACCP by the UK food industry

Implementation of HACCP by the UK food industry does vary across food sectors. The legal 'Own Checks' requirements and the associated approval

arrangements which apply to establishments manufacturing products of animal origin have resulted in a high proportion of these businesses implementing effective HACCP-based controls. This process has been influenced by the direct marketing benefits such systems offer businesses, particularly in the context of supplier requirements.

The implementation of HACCP or HACCP-based control systems in small and medium-sized enterprises (SMEs) – which represent the bulk of the over 600,000 food businesses operating in the UK – is more patchy. For these businesses, in the main retail and catering operations, there are significant practical barriers which have held back implementation rates.

Government funded studies have sought to identify these practical barriers to HACCP implementation.[3] Whilst barriers are likely to occur in any business regardless of size, they seem to be more acute in SMEs where the marketing benefits of introducing HACCP control systems are less clear. One UK survey[4] identified the following issues:

- *Cost*: HACCP was perceived as expensive, especially by smaller retailers and caterers. Such businesses were less able to identify any tangible benefits to balance out the costs of HACCP. Manufacturers on the other hand were more able to see the trading advantage HACCP gave them.
- *Time*: The more limited staffing and resources available in small businesses and the relative expense of external support meant that, in practice, HACCP implementation relied on managers putting in such systems themselves. This competed against many other jobs to be carried out.
- *Complexity*: HACCP was generally perceived as complex by managers of smaller businesses, particularly where many different products might be routinely handled, as in catering.
- *Perception of risk*: Interviews with managers, particularly in the retail and catering sector, indicated a generally low perception of the risks posed by their handling or preparation practices. This low perception of risk, even where establishments routinely handled both raw and ready-to-eat foods, influenced their motivation to introduce HACCP controls.

A recent WHO Expert Consultation examining HACCP implementation strategies for small and less developed food businesses (SLDBs) highlights the fact that these barriers are by no means restricted to the UK, and lists these and others which need to be overcome in any implementation strategy.[5] The consultation report put forward a number of measures to help address the barriers. These included:

- central production of sector specific generic HACCP guidance which could be readily tailored by the business
- reducing the complexity of HACCP plans and records by grouping together products with similar characteristics and which undergo similar processing steps
- the provision of accessible and relevant training on food hygiene and HACCP.

The regulatory nature of UK HACCP-based controls requirements means that in practice enforcers have an important role to play in overcoming these barriers and ensuring effective systems are in place. This is particularly important in SMEs, as local authority enforcement officials are frequently found to be the main sources of information and route for advice on food safety for these types of business. The UK HACCP implementation strategy, therefore, involves enforcement officials offering advice and support on HACCP as part of their routine enforcement approach. The nature of the advice necessary will vary across businesses, from those which have significant in-house technical resources and skills to draw on, to those businesses where their technical capability is very much smaller or absent altogether.

12.5 HACCP and enforcement

HACCP represented and still represents a culture change for many UK food businesses. Whilst most proprietors tend to accept the need for visually clean, well maintained premises, the concept of assessing risks and managing food safety in a more formalised way is not so readily assimilated alongside the many other competing commercial priorities.

HACCP not only imposes certain management disciplines, it also requires an underlying knowledge of food hygiene and safety that in some cases may necessitate additional training. Government codes of practice therefore advise enforcement authorities to adopt a graduated approach to the enforcement of HACCP.[6] As a first step towards securing compliance, authorised officers are asked to adopt an educative approach and discuss the legal requirements relating to HACCP, as well as training, with the proprietor. As such, attendance on a recognised food hygiene and safety course might be the first step toward a business putting HACCP-based management controls in place.

For businesses subject to product specific legislation, such as manufacturers of products of animal origin, effective HACCP-based controls would need to be in place for the business to be approved. However, in these businesses it is also more likely that in-house capabilities would reduce the reliance on external support.

The main purpose of the enforcement inspection is to establish whether food is being handled and processed hygienically and whether the food is safe to eat. Assessing the risks arising from the food business activities in the UK does, therefore, include an assessment of the effectiveness of any HACCP-based system and controls in place.

The inspection will also aim to identify any contraventions of food safety and hygiene legislation and seek to have them corrected through appropriate enforcement action or advice. In practice, these specific legal hygiene requirements which apply to premises structure, personnel and equipment form an important platform (prerequisite) for any HACCP control system.

Inspections will be part of a planned programme, e.g. as part of a six-month, yearly or longer inspection cycle, and include a thorough assessment of any relevant HACCP documentation, together with a visual examination of the premises, including checks on CCP controls. More targeted inspections or visits to establishments may result from a complaint or specific project or investigation. Such visits might also result in an assessment or partial assessment of the HACCP system.

Where food businesses have a documented food safety management system, officials are advised to vary their inspection approach and make use of the relevant material. Where there are satisfactory management controls as part of a well thought out HACCP system, and the authorised officer has confidence in the management of the business on the basis of previous inspections, the assessment of the in-house HACCP-based controls could be a significant part of the inspection and take up a major part of the time involved. The main purpose of any subsequent visual or physical examination would be to confirm that CCPs have been correctly identified and the necessary controls are in place. In some cases, on the basis of professional judgement, this could involve testing critical equipment or taking samples for independent analysis.

The need to assess HACCP has increased the focus of inspections on initial discussions with the proprietor or representative on the hygiene systems and procedures in place. This will include scrutiny of relevant documentation such as records relating to the monitoring of CCPs. As a result, inspections incorporating assessment of HACCP initially take significantly longer than previously was the case and may increase the need for follow-up visits. This has had an impact on overall numbers of inspections carried out in the UK, and this potential impact needs to be considered in any HACCP implementation strategy.

12.5.1 What will the enforcer be looking for?

A number of issues are listed below which enforcers may consider when assessing HACCP-based systems, and which might affect their subsequent questioning and lines of enquiry.

- *Confidence in the food business management controls*:
  - The track record of the company. Its willingness to act on previous advice and its enforcement and/or complaint history.
  - The attitude and activities of present management to hygiene issues, the level of in-house food hygiene knowledge and training.
  - Technical knowledge within, or accessible to the business.
  - The level of documentation and in-house food safety management controls, including verification procedures or any external third party quality assurance.
- *Adequacy of the HACCP arrangements*:

  - The scope of the HACCP system. Does it reflect the extent of business activities and take proper account of the nature of the product and its intended use?
  - Expertise – is sufficient in-house expertise present? Who developed the plan and control arrangements? Has appropriate use been made of external or generic guidance to ensure soundly based controls?
  - Flow chart – does it reflect practice? Is it routinely reviewed?
- *The hazard analysis*:
  - Have all reasonable hazards been identified?
  - Who completed the analysis?
  - Has due regard been given to any relevant industry guidance, expert report or published information relating to the sector?
- *Assessment of risk*:
  - What is the history of problems or complaints related to the premises? Officers may wish to see any records relating to complaints.
  - Severity and imminence of any hazards – are vulnerable customer groups supplied?
- *Assessment of controls*:
  - What are the specific controls in place?
  - Have all relevant critical controls been identified and are they operating to any recognised/legal standards, e.g. required temperatures?
  - Have critical limits been identified? Are they realistic, measurable and relevant to the process? How were they determined, are they sufficient and what monitoring takes place?
  - Are food handlers adequately instructed/trained, particularly where they are involved in any monitoring procedures? Are corrective actions sufficiently identified, understood and operated?
  - Are records maintained correctly? Enforcers might target an individual product or batch.
- *Verification procedures*:
  - Are there in-house checks that the HACCP system is working correctly?
  - What are the verification procedures?
  - Are they adequate?
  - Are documented HACCP procedures being followed and reviewed?
  - Are the necessary tools/equipment in place to ensure procedures can be followed or checks made?

12.5.2 How can you help prepare for an inspection?

Regulatory assessments of HACCP are likely to follow a similar pattern to any effective internal verification process. As such routine in-house monitoring will help provide much of the groundwork for any inspection. The following pointers should help facilitate inspections, providing benefit for both parties:

- *Records and documentation*: It is good practice to document the controls and keep records, even where this is not a legal requirement. Enforcement authorities may sometimes inform food businesses of inspections, particularly where HACCP systems are being assessed for the first time; however, many proceed on an unannounced basis. As such, keep documentation readily retrievable. Whether paper or electronic, the ability to access key documentation readily not only enables better management control, it will also oil the wheels of any inspection. Key documents will include:
  - the HACCP plan
  - significant procedures stemming from the plan
  - plans relating to layout and services
  - flow diagrams of the operation/operations
  - any monitoring records relating to identified CCPs
  - any verification activities.

  Documents need not be extensive where operations are not complex. In essence they should be sufficient for there to be confidence that controls are in place and maintained. In larger establishments, enforcers are unlikely in practice to be able to examine all documentation in the course of an inspection. They may, for example, select specific records relating to the day or a product and form a judgement on the system on that basis.
- *Training*: Records or procedures relating to training/instruction of staff on food safety should be readily to hand. Training in food hygiene and HACCP, particularly training associated with any nationally recognised course, will be viewed as an important component for an effective HACCP system.
- *Monitoring*: A significant focus of the inspection will be on the in-house monitoring arrangements, particularly those relating to the monitoring of CCPs. Demonstrating that routine in-house monitoring takes place and that staff have both the training and equipment to enable them to monitor effectively, will go some way in establishing that the legal requirements are being met.
- *In-house technical ability*: In-house staff who have particular food safety knowledge or responsibilities should be on hand where practical to answer questions and/or inform about the arrangements in place. These staff may be able to provide valuable information about the arrangements and the necessary controls which can help the enforcement process.

12.6 Common problems with HACCP systems

Listed below are some examples of common problems found with HACCP systems in businesses:

- *The HACCP system is too complicated*: This can be the case in smaller businesses where the driving force for the system has been largely reactive and in response to legislative requirements, rather than an evolving management process. Even in large manufacturers, supplier requirements

can be quite onerous and result in very complex systems not ideally suited to the business. Sometimes arrangements reflect off the shelf packages rather than the evolving needs and capabilities of the business itself. Simple systems and controls well understood and readily incorporated in the day-to-day operations and capabilities seem to be more successful and sustainable.

- *The HACCP system is not used*: Well prepared, documented manuals and procedures are present on site but not incorporated into day-to-day activities. This is relatively straightforward to see during any routine inspection and is often the consequence of overcomplicated, externally derived systems.
- *Too many critical control points identified*: There is a tendency to specify any hygiene control measure as critical. Whilst many operations need to be controlled and supervised on a routine basis, the number of CCPs, e.g. incorporating determination of critical limits, routine monitoring and corrective actions, should ideally be as few in number as possible. This cuts down on the relative complexity and degree of documentation necessary, and ensures resources are effectively focused.
- *Lack of monitoring*: Monitoring is crucial to effective control. As such, monitoring arrangements should be suited to the business processes and capabilities. Official inspections very often identify that routine monitoring of CCPs is not taking place, particularly where this is a manual process. Simple management checks and records will reduce the likelihood of this occurring.
- *Lack of corrective action*: Routine monitoring can become a non-thinking exercise. It is not unusual to examine business records, and see logs which indicate that critical limits have been breached, but no corrective action taken. Records are only a means to an end, not an end in themselves, and simple checks should be made to guard against this.
- *Poor hygiene*: Basic hygiene deficiencies within premises will undermine the HACCP system, which is why they are regarded as a prerequisite. For example, the increasing concern about microbiological hazards such as *E-coli* 0157 and *Listeria monocytogenes* make effective cleaning and disinfection procedures of paramount importance. Such prerequisite programmes, which are also basic legal requirements, can sometimes be found lacking alongside the HACCP 'system' which has assumed an almost mystical pre-eminence.

12.7 Future HACCP arrangements

The UK Food Standards Agency includes a remit to encourage the implementation of HACCP controls in the UK food industry. Sector specific initiatives will continue to be developed to accelerate HACCP take-up. For example, one recent centrally funded initiative in the retail butchery sector included the development of a generic HACCP manual, wall-chart and signage, free on-site advice and two days' food hygiene/HACCP training. This was provided to over 7000 butchers. Existing training programmes for enforcers and

industry are also being reviewed to identify further guidance material and tools to help more businesses put effective and practical food safety management systems in place.

On the European front, the EC recently issued a consolidated text on its proposals to fundamentally review the existing European food hygiene legislation. This contains a requirement for HACCP applicable to all European food businesses, with the exception of primary production. Future discussions to agree the text will no doubt draw from the experience of member states implementing the current HACCP-based regulatory requirements.

12.8 A personal perspective

There is currently a great deal of focus on the applicability of HACCP to smaller food businesses. Indeed within this chapter I have highlighted some of the difficulties faced by food businesses and enforcers. I believe that the take-up of HACCP in smaller operations has not been helped by HACCP's early association with larger manufacturers and its codification by Codex.[7] For example, its application in larger processors and the Codex Code of Practice guidance promotes the concept of teams of experts, and detailed specific product analysis. Whilst relevant for manufacturing establishments, it is clear that this approach is not well suited to smaller establishments handling large numbers of products, such as caterers.

Codex codification of HACCP has been very helpful in an international context. Indeed I have contributed to this process. However, this has probably served to restrict HACCP's evolution and further adaption to make it more suited to particular sector or business needs. This has helped to fuel concern and the perception that HACCP is too complex and ill suited to smaller food businesses.

Fortunately, international discussions are now taking place in the Codex Food Hygiene Committee which seem likely to result in the elaboration of further guidelines on HACCP application, particularly relevant to smaller non-manufacturing operations. These discussions reflect the conclusions and recommendations made in the WHO Expert Consultation Report and take account of the practical barriers to HACCP implementation in smaller operations.[5]

There is growing appreciation that HACCP, like any management system, needs to be flexible enough to take account of the practical needs of the business seeking to apply it. Whilst the seven principles provide an important framework to assist its common understanding and assessment, those principles should and can be applied in practice in different ways depending on the business type and complexity. There is a view, which I share, that this needs to be more formally recognised and described by Codex.

I believe HACCP is relevant to all types of food businesses. The responsibility for managing food safety and ensuring consumer protection rests

with the business, whatever its size. HACCP serves to make this responsibility clear. No amount of external inspection can compensate for inadequate day-to-day food safety awareness or management, and nor should it try. HACCP provides a logical way through which food safety management can be approached. It also provides a common framework to help proprietors, managers and enforcers in their assessment and subsequent confidence that operations are under routine control. The challenge (which in my view we have not yet met) is to ensure the HACCP-based controls, in small businesses in particular, are effective and proportionate and owned by the business.

It is my personal view that we are moving to second generation HACCP, which more effectively balances the practical needs of the business for a flexible and proportionate management system, with the needs for an internationally understood framework and approach. HACCP will be taken from the technocrats (like me), de-jargonised and given to the managers to adapt and truly incorporate into their business decision making. As someone recently said to me, 'HACCP is too important to be left to the experts!'

12.9 References

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